

Liouville–Hilbert basis conversion in `fc_pair_redfield.m`

Nucleus	Coordinates (x, y, z) (Å)	$\gamma/(2\pi)$ (MHz T ⁻¹)
¹⁹ F	(−3.9805, −0.5583, −0.6136)	40.0776
¹³ C	(−2.7608, −0.2372, −0.1625)	10.7084
¹ H	(−0.8183, +2.4463, +0.3685)	42.5774

Table 1: System parameters for the three-spin¹⁹F–¹³C–¹H model. Coordinates are from DFT optimisation of 4-fluorophenylalanine (final Standard-orientation block of the Gaussian log file). Gyromagnetic ratios are CODATA 2018 values. Spectral density for isotropic tumbling:

$$J(\omega) = \frac{\tau_c}{5(1 + \omega^2\tau_c^2)}, \quad \tau_c = 10^{-5} \text{ s.}$$

Scalar (coherent) couplings: $J(^{19}\text{F}–^{13}\text{C}) = 243.5 \text{ Hz}$, $J(^{13}\text{C}–^1\text{H}) = 10.71 \text{ Hz}$, $J(^{19}\text{F}–^1\text{H}) = 0$.

List of Zeeman tensors:

For ¹⁹F: [−113.0955 −13.4634 23.3880, −13.4634 −32.0922 −17.6116, 23.3880 −17.6116 −153.6685]

For ¹³C: [240.8318 25.3402 53.6419, 25.3402 162.9404 5.1940, 53.6419 5.1940 127.8480]

For ¹H: [5.0782 2.5669 −1.6492, 2.5669 10.7485 0.0000, −1.6492 0.0000 11.9705]

1 Two representations of the same physics

A two-spin- $\frac{1}{2}$ system (here ¹⁹F–¹³C) has a four-dimensional complex Hilbert space $\mathcal{H} \cong \mathbb{C}^4$ spanned by the Zeeman product states $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$. A density matrix ρ is a 4×4 Hermitian, positive-semidefinite operator on $\mathcal{H}$ with unit trace.

The Redfield master equation is most naturally written as a superoperator equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \mathcal{R}(\rho), \tag{1}$$

where $\mathcal{R}$ is a linear map from 4×4 matrices to 4×4 matrices. To handle $\mathcal{R}$ as an ordinary matrix–vector product, one “vectorises” ρ into a 16-dimensional column vector, which is

the *Liouville space* representation. Spinach uses this internally, and all quantities returned by `hamiltonian()` and `relaxation()` live in this space.

2 The product operator basis

Let $\{B_k\}_{k=1}^{16}$ be the set of two-spin product operators built from single-spin operators

$$\mathbf{1}, \quad I_z, \quad I_+, \quad I_- \quad (2)$$

for each of the two spins. Explicitly, using the tensor-product notation $A \otimes B \equiv \text{kron}(A, B)$:

$$\{B_k\} = \{\mathbf{1}^{(1)} \otimes \mathbf{1}^{(2)}, I_z^{(1)} \otimes \mathbf{1}^{(2)}, \mathbf{1}^{(1)} \otimes I_z^{(2)}, I_+^{(1)} \otimes \mathbf{1}^{(2)}, \dots, I_-^{(1)} \otimes I_-^{(2)}\}, \quad (3)$$

where all 16 pairwise combinations of the four single-spin operators are included. These operators are 4×4 matrices; they span the space of all 4×4 complex matrices and thus form a basis for Liouville space.

NMR (unnormalised) convention. Spinach uses the *unnormalised* product operator convention: the identity on both spins is $B_1 = \mathbf{1}_4$, the 4×4 identity matrix. In particular $\|B_1\|_F = \sqrt{\text{Tr}(\mathbf{1}_4)} = 2 \neq 1$. This differs from the Frobenius-orthonormal IST basis sometimes used in the literature.

3 The Spinach Liouville vector

In the `sphten-liouv` formalism Spinach stores the density matrix as a coefficient vector

$$\vec{\rho} \in \mathbb{C}^{N_L}, \quad N_L = 16, \quad (4)$$

where $\vec{\rho}_k$ is the coefficient of B_k in the expansion of ρ :

$$\rho = \sum_{k=1}^{N_L} \vec{\rho}_k B_k. \quad (5)$$

Spinach's `state(spin_system, op, spin)` function returns a sparse column vector `s` with a single dominant entry at the position j corresponding to the product operator matching `op` on `spin`. Hence the Liouville vector $\vec{\rho}_0$ for the initial state $\rho_0 = |\uparrow\downarrow\rangle\langle\uparrow\downarrow|$ is constructed by decomposing in the product operator basis:

$$\begin{aligned} \rho_0 &= |\uparrow\rangle\langle\uparrow|_1 \otimes |\downarrow\rangle\langle\downarrow|_2 = \left(\frac{1}{2} + I_z^{(1)}\right) \otimes \left(\frac{1}{2} - I_z^{(2)}\right) \\ &= \frac{1}{4} \mathbf{1}_4 + \frac{1}{2} I_z^{(1)} \otimes \mathbf{1} - \frac{1}{2} \mathbf{1} \otimes I_z^{(2)} - I_z^{(1)} \otimes I_z^{(2)}, \end{aligned} \quad (6)$$

from which the Liouville vector is read off coefficient by coefficient. This is directly what the script does:

```

rho0 = (1/4)*state(spin_system,'E','all') ...
+ (1/2)*state(spin_system,'Lz',1) ...
- (1/2)*state(spin_system,'Lz',2) ...
- state(spin_system',{'Lz','Lz'},{1,2});

```

4 The conversion matrix T

Define the $16 \times N_L$ matrix T by

$$T_{:,k} = \text{vec}(B_k), \quad (7)$$

where $\text{vec}(M)$ stacks the columns of a matrix M into a column vector (standard MATLAB column-major ordering). Then, from (5):

$$\text{vec}(\rho) = T \vec{\rho}, \quad (8)$$

or equivalently, $\rho = \text{reshape}(T \vec{\rho}, 4, 4)$. This is exactly what `liouv2hilb` computes:

```

liouv2hilb = @(v) reshape(T_liouv2hilb * full(v), 4, 4);

```

Building T in the code. Because Spinach may scale its basis vectors internally, we cannot simply identify $\text{vec}(B_k)$ with the k -th standard basis vector. Instead, for each product operator B_r in `prod_ops`:

1. call `sv = state(spin_system, label_r, spin_r)`, obtaining a sparse vector with a single dominant entry $\text{sv}_j \neq 0$ at position $j = \arg \max_i |\text{sv}_i|$;
2. set $T_{:,j} = \text{vec}(B_r) / \text{sv}_j$.

The division by sv_j absorbs Spinach's normalisation so that (8) holds. To see why, note that the Spinach propagator advances $\vec{\rho}$ so that $\vec{\rho}_j$ accumulates the coefficient of B_r according to Spinach's internal normalisation; dividing by sv_j converts back to the physical expansion (5).

Consistency check for ρ_0 . One can verify (8) by substituting the `rho0` vector into `reshape(T \vec{\rho}_0, 4, 4)` and checking it equals the explicit 4×4 matrix with a single 1 at position (2,2), corresponding to the $|\uparrow\downarrow\rangle$ state in the Zeeman ordering $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$.

5 Population extraction and why $T^\top \neq T^{-1}$

The population of eigenstate $|n\rangle$ at time t is

$$P_n(t) = \text{Tr}(|n\rangle \langle n| \rho(t)) = \text{vec}(|n\rangle \langle n|)^\top \text{vec}(\rho(t)). \quad (9)$$

Using (8) to substitute $\text{vec}(\rho(t)) = T \vec{\rho}(t)$:

$$P_n(t) = \text{vec}(|n\rangle \langle n|)^\top T \vec{\rho}(t) = \underbrace{(T^\top \text{vec}(|n\rangle \langle n|))^\top}_{=: \mathbf{p}_n^\top} \vec{\rho}(t). \quad (10)$$

The Liouville co-vector $\mathbf{p}_n = T^\top \text{vec}(|n\rangle \langle n|)$ is computed in the script as:

```
rho_n = V(:,n)*V(:,n)'; % 4x4 projector |n><n|
proj_liouv(:,n) = T_liouv2hilb' * rho_n(:);
```

Why not T^{-1} ? One might naively think of the projector as the “inverse image” of $|n\rangle \langle n|$ under T and use $T^{-1} \text{vec}(|n\rangle \langle n|)$. However, the correct formula (10) involves $T^\top$, not T^{-1} . These coincide only when T is unitary, i.e. when the basis $\{B_k\}$ is Frobenius-orthonormal:

$$\text{Tr}(B_j^\dagger B_k) = \delta_{jk}. \quad (11)$$

Spinach’s product operator basis does *not* satisfy this: for example,

$$\|\mathbf{1}_4\|_F^2 = \text{Tr}(\mathbf{1}_4) = 4 \neq 1, \quad (12)$$

so T is not unitary and $T^\top \neq T^{-1}$. Using T^{-1} for the projector would give incorrect populations at $t = 0$ (not equal to $1/2$ for the singlet and T_0 components of $|\uparrow\downarrow\rangle$ at $B_0 = 0$, as required by the decomposition $|\uparrow\downarrow\rangle = (|T_0\rangle + |S_0\rangle)/\sqrt{2}$).

Trace normalisation. An equivalent expression for the population is

$$P_n(t) = \text{Tr}(|n\rangle \langle n| \rho(t)) = \text{vec}(|n\rangle \langle n|)^\top T \vec{\rho}(t), \quad (13)$$

which also immediately gives the trace of ρ as

$$\text{Tr} \rho(t) = \text{vec}(\mathbf{1}_4)^\top T \vec{\rho}(t) = \sum_{k=1}^{16} \vec{\rho}_k(t) \text{Tr}(B_k). \quad (14)$$

Since only $B_1 = \mathbf{1}_4$ has a non-zero trace ($\text{Tr}(\mathbf{1}_4) = 4$), this reduces to $\text{Tr} \rho = 4 \vec{\rho}_1$, confirming that the identity coefficient is fixed by trace normalisation.

6 Summary of the data flow in the script

1. **Spinach setup:** `create`, `basis`, `hamiltonian`, `relaxation` produce the Liouville-space generator $G = -iH_L + R$ (both 16×16 , where H_L is the commutator super-operator of H).
2. **Initial state:** $\vec{\rho}_0$ is assembled from `state()` calls using the decomposition (6).
3. **Conversion matrix T :** built column-by-column via `state()` index probing; satisfies $\text{reshape}(T\vec{\rho}, 4, 4) = \rho$.
4. **Eigenbasis:** H is diagonalised as a 4×4 Hilbert-space matrix (built from `kron` products) to obtain eigenstates $|n\rangle$.
5. **Co-vectors:** $\mathbf{p}_n = T^\top \text{vec}(|n\rangle \langle n|)$ are precomputed once.
6. **Propagation:** $\vec{\rho}(t + \delta t) = e^{G\delta t} \vec{\rho}(t)$; populations $P_n(t) = \mathbf{p}_n^\top \vec{\rho}(t)$.
7. **Positivity check:** $\rho(t) = \text{reshape}(T\vec{\rho}(t), 4, 4)$; smallest eigenvalue of $\rho(t)$ flags breakdown.

7 Singlet protection in two spins and its breakdown at three spins

7.1 Why S is conserved in the two-spin ZULF Hamiltonian

At zero field ($B_0 = 0$) the two-spin Hamiltonian reduces to

$$H = J \mathbf{I}_1 \cdot \mathbf{I}_2 = \frac{J}{2} (S_{\text{tot}}^2 - \mathbf{I}_1^2 - \mathbf{I}_2^2), \quad (15)$$

which is proportional to S_{tot}^2 up to a constant. Hence $[H, S_{\text{tot}}^2] = 0$: the total spin quantum number S is exactly conserved by the Hamiltonian, and the singlet ($S = 0$) and triplet ($S = 1$) manifolds are dynamically decoupled.

7.2 The singlet is a dark state for DD relaxation

Redfield relaxation from isotropic molecular tumbling is built from the rank-2 spherical tensor components $T_q^{(2)}(\mathbf{I}_1, \mathbf{I}_2)$ of the dipole–dipole (DD) Hamiltonian. One can show algebraically that every component annihilates the singlet:

$$T_q^{(2)}(\mathbf{I}_1, \mathbf{I}_2) |S=0, M=0\rangle = 0 \quad \forall q. \quad (16)$$

Three cases make this explicit:

- $q = 0$: $T_0^{(2)} \propto 3I_{z1}I_{z2} - \mathbf{I}_1 \cdot \mathbf{I}_2$. In the singlet, $\mathbf{I}_1 \cdot \mathbf{I}_2 = -\frac{3}{4}$ and $I_{z1}I_{z2} = -\frac{1}{4}$, giving $3(-\frac{1}{4}) - (-\frac{3}{4}) = 0$.
- $q = \pm 1$: $T_{\pm 1}^{(2)} \propto I_{\pm 1}I_{z2} + I_{z1}I_{\pm 2}$. Applying to $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ produces two terms that cancel exactly due to the antisymmetry of the singlet.
- $q = \pm 2$: $T_{\pm 2}^{(2)} \propto I_{\pm 1}I_{\pm 2}$, which requires both spins to flip simultaneously; acting on the singlet yields zero because $I_+|\uparrow\rangle = 0$.

At zero field the CSA contribution to $\mathcal{R}$ also vanishes (it scales as B_0^2). Consequently the Redfield superoperator $\mathcal{R}$ cannot transfer population into or out of the singlet sector.

The singlet is therefore *doubly protected*: both the Hamiltonian and the dissipator preserve S . Starting from $|\uparrow\uparrow\rangle = |S=1, M=+1\rangle$, only the three triplet states can exchange population; the singlet remains exactly zero. This is the physical origin of the nuclear spin long-lived singlet state.

7.3 Why the third spin breaks the protection

Consider extending the system to three spins with Hamiltonian

$$H = J_{12} \mathbf{I}_1 \cdot \mathbf{I}_2 + J_{23} \mathbf{I}_2 \cdot \mathbf{I}_3. \quad (17)$$

Using the commutator identity $[\mathbf{I}_j \cdot \mathbf{I}_k, \mathbf{I}_j \cdot \mathbf{I}_l] = i \mathbf{I}_j \cdot (\mathbf{I}_k \times \mathbf{I}_l)$, one finds that each individual term commutes with S_{tot}^2 : $[J_{jk} \mathbf{I}_j \cdot \mathbf{I}_k, S_{\text{tot}}^2] = 0$. The Hamiltonian therefore still conserves total spin, so the $S = 3/2$ and $S = 1/2$ sectors remain decoupled under unitary evolution even when $J_{12} \neq J_{23}$.

The breakdown comes entirely from the **relaxation superoperator**. The three-spin Hilbert space is conveniently classified by the subsystem spin S_{12} of the (1,2) pair:

$$\begin{aligned} S_{12} = 0 \otimes I_3 = \frac{1}{2} : S_{\text{tot}} = \frac{1}{2} \quad [\text{doublet (2)}] \\ S_{12} = 1 \otimes I_3 = \frac{1}{2} : S_{\text{tot}} = \frac{1}{2} \text{ or } \frac{3}{2} \quad [\text{doublet (1) and quartet}] \end{aligned} \quad (18)$$

The pair operator $T_q^{(2)}(\mathbf{I}_1, \mathbf{I}_2)$ still annihilates the $S_{12} = 0$ component by (16), but it *does* act on the $S_{12} = 1$ component and can induce the transition $S_{12} = 1 \rightarrow S_{12} = 0$. Via Clebsch–Gordan recoupling with the spectator spin I_3 , this maps

$$|S_{\text{tot}} = \frac{3}{2}, M\rangle \xrightarrow{T_q^{(2)}(\mathbf{I}_1, \mathbf{I}_2)} |S_{12} = 0\rangle \otimes |I_3, m_3\rangle = |S_{\text{tot}} = \frac{1}{2}, \text{doublet (2), } M'\rangle, \quad (19)$$

directly mixing the two total-spin sectors. Analogous arguments apply to the pairs (1,3) and (2,3); all three DD interactions participate in crossing the S_{tot} boundary.

The essential difference is summarised in Table 2.

System	S_{tot} conserved by H ?	S_{tot} conserved by $\mathcal{R}$?
2-spin, ZULF	✓	✓
3-spin, ZULF	✓	×

Table 2: Conservation of total spin under Hamiltonian and Redfield relaxation.

The key physical intuition is that in the two-spin case the dark-state condition (16) is an *exact* property of the full relaxation superoperator, not just of the Hamiltonian. The third spin acts as a spectator that converts pair-level singlet–triplet flips ($\Delta S_{12} = 1$) into total-spin-changing transitions ($\Delta S_{\text{tot}} = 1$), unlocking population flow between the $S = 1/2$ doublets and the $S = 3/2$ quartet. This is also why the Redfield equation is more prone to positivity violations in larger spin systems: the additional cross-manifold pathways opened by the spectator spin feed population into coherences that the Born–Markov approximation cannot track accurately.

8 Density matrix positivity breakdown

The Hamiltonian of a target system (H_S) coupled to a bath (with Hamiltonian H_B) is given by

$$H = H_S + H_B + H_{SB} \quad (20)$$

where

$$H_{SB} = \sum_{\alpha} A_{\alpha} B_{\alpha} \quad (21)$$

Consider the general form of a Redfield equation

$$\dot{\rho} = -i[H, \rho] + \sum_{i,j} \left(-\frac{K_{i,j}^+(\omega_j) - K_{i,j}^-(\omega_i)}{2} [\sigma_i^{\dagger} \sigma_j, \rho] + \frac{K_{i,j}^+(\omega_j) + K_{i,j}^-(\omega_i)}{2} \left(-\{\sigma_i^{\dagger} \sigma_j, \rho\} + 2\sigma_j \rho \sigma_i^{\dagger} \right) \right) \quad (22)$$

where $K_{ij}^+(\omega_j) = \Gamma_{ij}(\omega_j)/2 \pm i\Lambda_{ij}(\omega_j)$, in which both $\Gamma_{ij}(\omega_j)$ and $\Lambda_{ij}(\omega_j)$ can be understood as the matrix of dephasing rates (energy renormalizations) integrated over the weights of transitions i,j in the coordinates of the system coupled to the bath:

$$\begin{aligned} \Gamma_{ij}(\omega) &= \sum_{\alpha,\beta} (A_{\alpha})_i^* \gamma_{\alpha\beta}(\omega) (A_{\beta})_j \\ \Lambda_{ij}(\omega) &= \sum_{\alpha,\beta} (A_{\alpha})_i^* \lambda_{\alpha\beta}(\omega) (A_{\beta})_j \end{aligned}$$

where $(A_{\beta})_j = \text{Tr}(\sigma_j^{\dagger} A_{\beta})$.

In the zero-field limit, we note that the dynamics for a pair of spins coupled through

Heisenberg interactions can be described effectively in terms of a single zero transition frequency, which is a consequence of the energetic degeneracy in the triplet manifold and that the singlet state spans the null space of the rank-2 spherical tensor components of the dipole-dipole Hamiltonian, as described above. Thus, there is no possibility of asymmetry $\Gamma_{ij}(\omega_j) \neq \Gamma_{ij}(\omega_i)$ which is one condition to break Lindbladian evolution. In fact, we get positive density matrices across the evolution for our simple two-spin model for arbitrary values of J coupling and for relatively low solvent correlation times.

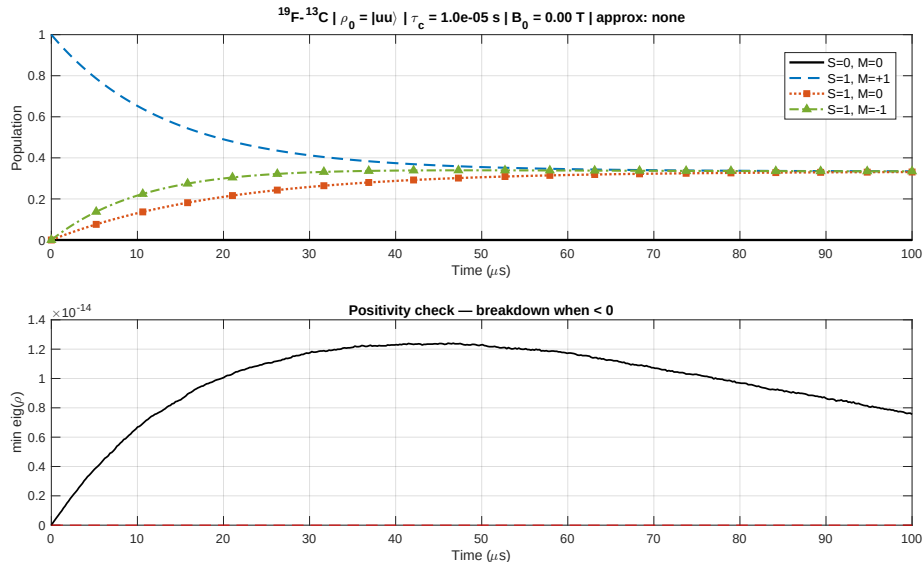


Figure 1: Upper panel: evolution of population in the Hamiltonian eigenbasis for two Heisenberg coupled spins with strength $J = 10^9$ Hz at zero field and considering $|\uparrow\uparrow\rangle$ as the initial state. Lower panel: lowest density matrix eigenvalue tracking across the timescale of time evolution.

On the other hand, we get a different picture when extending the model to 3 interacting spins through Heisenberg interactions. By choosing a J coupling on the same order of magnitude as τ_c^{-1} (which corresponds to the width of the spectral density) we might obtain unphysical density matrices across the evolution:

9 Linbladian Master equation

From the previous intuition, we conclude that given the mismatch between the time-scale of the bath correlation time and the isotropic J couplings in the zero-field limit, density matrix positivity is preserved and at the same time a Lindbladian form of the master equation is

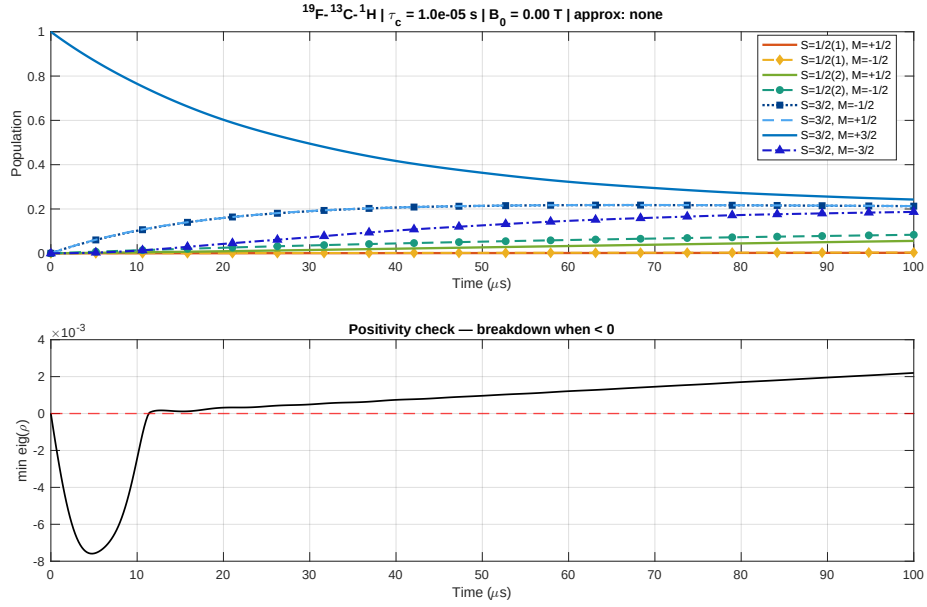


Figure 2: Upper panel: evolution of population in the Hamiltonian eigenbasis for three Heisenberg coupled spins (according to the connectivity taken from a reference molecule) with strength $J = 10^5$ Hz at zero field and considering $|\uparrow\uparrow\uparrow\rangle$ as the initial state. Lower panel: lowest density matrix eigenvalue tracking across the timescale of time evolution.

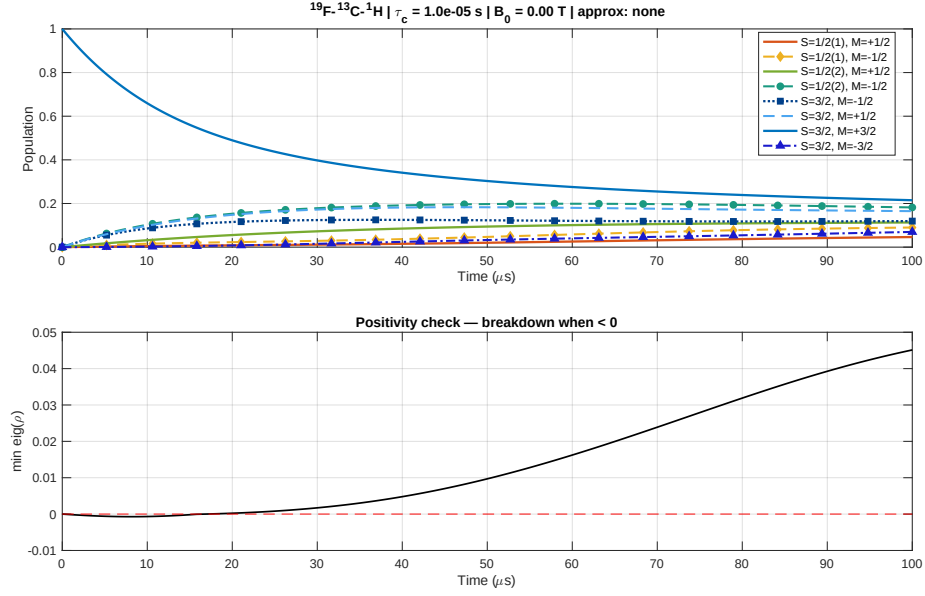


Figure 3: Upper panel: evolution of population in the Hamiltonian eigenbasis for three Heisenberg coupled spins (according to the connectivity taken from a reference molecule) with strength $J = 10^4$ Hz at zero field and considering $|\uparrow\uparrow\uparrow\rangle$ as the initial state. Lower panel: lowest density matrix eigenvalue tracking across the timescale of time evolution.

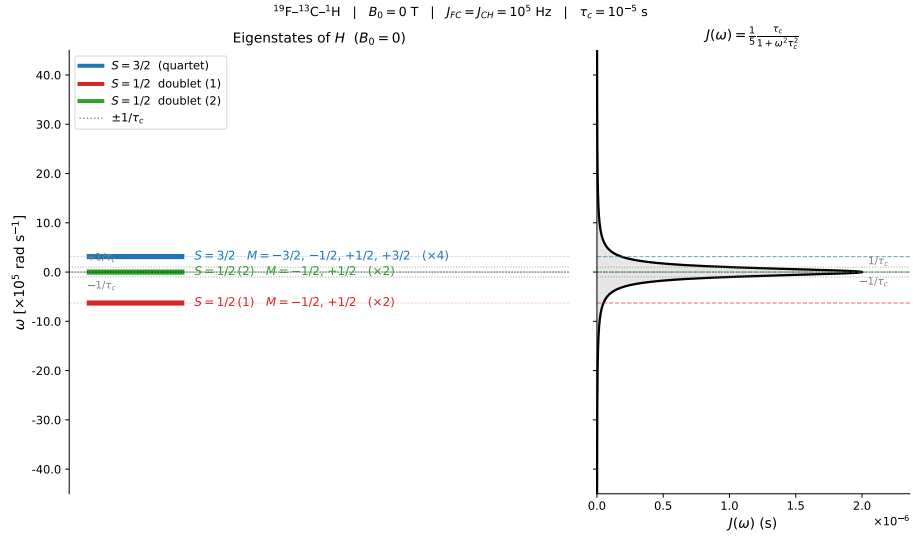


Figure 4: Comparison between the spectrum of three strongly coupled spins and the spectral density of the bath, choosing parameters that break density matrix positivity during evolution, *i.e.* $J \sim 1/\tau_c$

warranted. For our toy model we have the Hamiltonian

$$H = \underbrace{\sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j}_{H_0} + H_{DD}(t)$$

where $H_{DD}(t)$ is the time-modulated dipolar Hamiltonian, which in an Irreducible Spherical Tensor (IST) basis can be written as

$$H_{DD}(t) = \sum_{k,m=-2}^2 D_{k,m}^{(2)}(\Omega(t)) \hat{Q}_{k,m}$$

where

$$\hat{Q}_{k,m} = \sum_{i,j} \left(-\frac{\mu_0}{4\pi} \frac{\gamma_i \gamma_j}{r_{ij}^3} a_{2,m}^{ij} T_{2,k}^{i,j} \right)$$

$D_{k,m}^{(2)}(\Omega(t))$ being rank-2 Wigner functions time-modulated by the function $\Omega(t)$, $T_{2,k}^{i,j}$ being the rank-2 two-body IST operators for the dipolarly connected spins i, j ; $a_{2,m}^{ij}$ are orientational factors:

$$a_{2,\pm 2}^{ij} = \frac{1}{2} (a_{x,x}^{ij} - a_{y,y}^{ij} \mp i (a_{x,y}^{ij} + a_{y,x}^{ij})) \quad (23)$$

$$a_{2,\pm 1}^{ij} = \mp \frac{1}{2} (a_{x,z}^{ij} + a_{z,x}^{ij} \mp i (a_{y,z}^{ij} + a_{z,y}^{ij})) \quad (24)$$

$$a_{2,0}^{ij} = \frac{1}{\sqrt{6}} (2a_{z,z}^{ij} - (a_{x,x}^{ij} + a_{y,y}^{ij})) . \quad (25)$$

which are symmetrized linear combinations of elements of the dipolar tensor $\mathbf{A}$ ($a_{\alpha,\beta}^{ij} = \mathbf{e}_\alpha \cdot \mathbf{A}^{ij} \cdot \mathbf{e}_\beta$)

$$H_{DD} = \sum_{i,j} \left(-\frac{\mu_0}{4\pi} \frac{\gamma_i \gamma_j}{r_{ij}^3} \right) \mathbf{S}_i \cdot \mathbf{A}^{ij} \cdot \mathbf{S}_j,$$

and the rest of the prefactors encode the distance-dependent strength of the dipolar interaction. From de la Pradilla's prescription to build a Redfield equation in the eigenbasis of H_0 , it suffices to identify that the bath degrees of freedom are implicitly encoded in the Wigner functions $D_{k,m}^{(2)}(\Omega(t))$. Ignoring for the time being Lamb-shifts, we have, for isotropic molecules, that (22)

$$\begin{aligned} \frac{\gamma_{\alpha,\beta}(\omega)}{2} &= \int_0^\infty d\tau \langle D_\alpha^{(2)}(\tau) D_\beta^{(2)}(0) \rangle e^{i\omega\tau} \\ &= \delta_{\alpha,\beta} J(\omega) \end{aligned}$$

where we introduce the composite indices $\alpha = (k, m)$, $\beta = (k', m')$. From which it follows that

$$\begin{aligned}\Gamma_{ij}(\omega) &= \sum_{\alpha\beta} (A_\alpha)_i^* \gamma_{\alpha\beta}(\omega) (A_\beta)_j \\ &= 2J(\omega) \sum_{\alpha} (\hat{Q}_\alpha)_i^* (\hat{Q}_\alpha)_j\end{aligned}$$

and $(\hat{Q}_\alpha)_j = \text{Tr}\{\sigma_j^\dagger \hat{Q}_\alpha\}$. Plugging this in Eq. (22) we get

$$\begin{aligned}\dot{\rho} &= -i[H, \rho] - \frac{1}{4} \sum_{i,j} (\Gamma_{ij}(\omega_j) - \Gamma_{ij}(\omega_i)) [\sigma_i^\dagger \sigma_j, \rho] \\ &\quad + \frac{1}{4} \sum_{i,j} (\Gamma_{ij}(\omega_j) + \Gamma_{ij}(\omega_i)) \left(-\{\sigma_i^\dagger \sigma_j, \rho\} + 2\sigma_j \rho \sigma_i^\dagger \right)\end{aligned}\tag{26}$$

where for completeness in this section $\sigma_i = |n_i\rangle\langle m_i|, |n_i\rangle, |m_i\rangle$ being eigenstates of H_0 , and $\omega_j = E_{m_j} - E_{n_j}$ is the energy for transition j . One immediate question here, to understand the selection rules for transitions between H_0 eigenstates, is how the symmetry-adapted observables $\hat{Q}_{k,m}$ act on the eigenstates of the isotropic Hamiltonian H_0 . More precisely, for

$$\hat{Q}_{k,m}|J, M\rangle = \sum_{K,L} c_{KL}^{(k,m)} |K, L\rangle$$

the question is to find which amplitudes $c_{KL}^{(k,m)}$ are not vanishing.

9.1 Fermi's Golden Rule

Getting some intuition on what are the dominant transition rates across populations in the dynamics in the low-field limit. Taking as our starting point the time domain formulation of FGR:

$$k_{i \rightarrow f}(\omega) = \int_0^\infty d\tau \langle V_{fi}(t) V_{fi}^\dagger(0) \rangle$$

where we make the identification $V \rightarrow H_{DD}(t) = \sum_{k,m} D_{k,m}(t) Q_{k,m}$. To get

$$k_{i \rightarrow f} = \sum_{k,m} \sum_{k',m'} \int_0^\infty d\tau e^{i\omega_{fi}\tau} \langle D_{k,m}(\tau) D_{k',m'}(0) \rangle \langle f | Q_{k,m} | i \rangle \langle i | Q_{k',m'} | f \rangle$$

where we considered $|i\rangle, |f\rangle$ being eigenstates of the spin Hamiltonian, $\omega_{fi} = \omega_f - \omega_i$ where ω_j is the eigenfrequency of the j th eigenstate. For isotropic tumbling we have

$$k_{i \rightarrow f}(\omega) = J(\omega_{fi}) \sum_{k,m} |\langle f | Q_{k,m} | i \rangle|^2$$

Rank	Initial $ i\rangle$	Final $ f\rangle$	$\omega_{fi}/(2\pi)$ [Hz]	$k/(2\pi)$ [Hz]
1	$ 1/2_\beta, -1/2\rangle$	$ 3/2, +3/2\rangle$	+7.94	1603
2	$ 3/2, +3/2\rangle$	$ 1/2_\beta, -1/2\rangle$	-7.94	1603
3	$ 3/2, -3/2\rangle$	$ 1/2_\beta, +1/2\rangle$	-7.94	1603
4	$ 1/2_\beta, +1/2\rangle$	$ 3/2, -3/2\rangle$	+7.94	1603
5	$ 3/2, +1/2\rangle$	$ 1/2_\beta, -1/2\rangle$	-7.94	1202

Table 3: Top-5 Fermi's Golden Rule population-transfer rates $k_{i \rightarrow f} = J(\omega_{fi}) \sum_{k,m} |\langle f | Q_{k,m} | i \rangle|^2$ between eigenstates of H_0 , computed for $\tau_c = 10^{-5}$ s, $B_0 = 0$. States $|1/2_\beta, M\rangle$ are the higher-energy $S = 1/2$ doublet; $|3/2, M\rangle$ are the quartet states. Rates 1–4 are the two time-reversal pairs at the $S = 3/2 \leftrightarrow S = 1/2_\beta$ gap $|\omega_{fi}|/(2\pi) \approx 7.94$ Hz; rate 5 involves a different ΔM channel at the same gap. All transitions are forbidden within the doublet sector, consistent with the triangle rule.

Table 3 lists the five largest rates computed numerically for $\tau_c = 10^{-5}$ s and $B_0 = 0$, using the physical geometry and couplings of the $^{19}\text{F}-^{13}\text{C}-^1\text{H}$ system.

Table 4 restricts the view to the initial state $|3/2, +3/2\rangle = |\uparrow\uparrow\uparrow\rangle$, exposing the outflow channels that govern the early-time dynamics.

Rank	Final $ f\rangle$	$\omega_{fi}/(2\pi)$ [Hz]	$k/(2\pi)$ [Hz]
1	$ 1/2_\beta, -1/2\rangle$	-7.94	1603
2	$ 3/2, +1/2\rangle$	0	1077
3	$ 3/2, -1/2\rangle$	0	1077
4	$ 1/2_\beta, +1/2\rangle$	-7.94	401
5	$ 1/2_\alpha, -1/2\rangle$	-246.27	1.29

Table 4: Top-5 FGR transfer rates out of $|3/2, +3/2\rangle = |\uparrow\uparrow\uparrow\rangle$, for $\tau_c = 10^{-5}$ s, $B_0 = 0$. Ranks 2–3 are intra-quartet ($\omega_{fi} = 0$, driven by $\Delta M = -1$ and $\Delta M = -2$ components of $Q_{k,m}$); ranks 1 and 4 cross into $S = 1/2_\beta$ at the 7.94 Hz gap. Rank 5, reaching the lower-energy $S = 1/2_\alpha$ doublet at a 246 Hz gap, is three orders of magnitude slower because the α doublet has predominantly pair-singlet character and therefore a small reduced matrix element. No intra-doublet transitions appear, consistent with the triangle-rule selection rules.

9.2 Relaxation matrix verification

Reproduction of the reference Redfield dynamics through Spinach package remains an elusive task. Comparison between the Redfield relaxation matrix (in the Liouville Pauli basis) and the Linbladian one flags an almost two-order-of-magnitude mismatch between 3 diagonal matrix elements. Closer inspection of this mismatch might reveal the root of the issues.

At its core, in terms of Hamiltonian and relaxation super-operators we have

$$\partial_t \rho = -i\mathcal{H}_0 \rho + \mathcal{R} \rho$$

where

$$\mathcal{R} = - \sum_{\alpha, \beta} \int_0^\infty d\tau g_{\alpha, \beta}(\tau) \mathcal{Q}_\alpha e^{-i\mathcal{H}_0 \tau} \mathcal{Q}_\beta^\dagger e^{i\mathcal{H}_0 \tau}$$

and we used the composite indices $\alpha = k, m$. Decomposing the super-operators in terms of the $\mathcal{H}_0$ eigenbasis we have

$$\mathcal{R} = - \sum_{\alpha, \beta} \sum_{i, j} \int_0^\infty d\tau g_{\alpha, \beta}(\tau) e^{-i\omega_j \tau} c_i^\alpha c_j^{\beta*} \mathcal{P}_i \mathcal{P}_j^\dagger$$

where the composite indices i, j label transition operators of the form $|n_j\rangle\langle m_j|$ with a corresponding transition frequency $\omega_j = \omega_{n_j} - \omega_{m_j}$, $H_0|n_j\rangle = \omega_{n_j}|n_j\rangle$, and $c_i^\alpha = \text{Tr}\{\sigma_i^\dagger \mathcal{Q}_\alpha\}$. To make the connection with the Redfield form above we identify $\Gamma_{ij}(\omega) = \sum_{\alpha, \beta} \int_0^\infty d\tau g_{\alpha, \beta}(\tau) e^{-i\omega \tau} c_i^\alpha c_j^{\beta*}$, such that

$$\begin{aligned} \mathcal{R}[\rho] &= - \sum_{i, j} \Gamma_{ij}(\omega_j) [\sigma_i, [\sigma_j^\dagger, \rho]] \\ &= - \sum_{i, j} \Gamma_{ij}(\omega_j) [\sigma_i^\dagger, [\sigma_j, \rho]] \\ &= -\frac{1}{2} \sum_{i, j} \Gamma_{ij}(\omega_j) [\sigma_i^\dagger, [\sigma_j, \rho]] - \frac{1}{2} \sum_{i, j} \Gamma_{ij}(\omega_i) [\sigma_j, [\sigma_i^\dagger, \rho]] \\ &= -\frac{1}{2} \sum_{i, j} \Gamma_{ij}(\omega_j) \left(\sigma_i^\dagger \sigma_j \rho - \sigma_i^\dagger \rho \sigma_j - \sigma_j \rho \sigma_i^\dagger + \rho \sigma_j \sigma_i^\dagger \right) \\ &\quad - \frac{1}{2} \sum_{i, j} \Gamma_{ij}(\omega_i) \left(\sigma_j \sigma_i^\dagger \rho - \sigma_j \rho \sigma_i^\dagger - \sigma_i^\dagger \rho \sigma_j + \rho \sigma_i^\dagger \sigma_j \right) \\ &= - \sum_{ij} \frac{\Gamma_{ij}(\omega_i) + \Gamma_{ij}(\omega_j)}{2} \{ \sigma_i^\dagger \sigma_j, \rho \} - \sum_{ij} (\Gamma_{ij}(\omega_i) + \Gamma_{ij}(\omega_j)) \sigma_i^\dagger \rho \sigma_j \\ &\quad + \sum_{i, j} \frac{\Gamma_{ij}(\omega_i) - \Gamma_{ij}(\omega_j)}{2} [\sigma_i^\dagger \sigma_j, \rho] \end{aligned}$$

9.3 Core tool: Wigner–Eckart theorem

The key structural observation is that $T_{2,k}^{i,j}$, and therefore $\hat{Q}_{k,m}$ (for fixed m , as a function of k), is a *rank-2 irreducible spherical tensor operator under global rotations of all spins*. This follows because $T_{2,k}^{i,j}$ is a bilinear combination of single-spin operators that transforms as $D^{(2)}$ under $e^{i\theta\mathbf{n}\cdot\mathbf{J}_{\text{tot}}}$; the fact that it only has support on the pair (i, j) is irrelevant for its transformation properties under global rotations. Since H_0 is rotationally invariant, its eigenstates $|J, M\rangle$ carry good total angular momentum, and the Wigner–Eckart theorem gives

$$\langle K, L, \nu' | \hat{Q}_{k,m} | J, M, \nu \rangle = \langle J, M; 2, k | K, L \rangle \langle K, \nu' | \hat{Q}_m | J, \nu \rangle, \quad (27)$$

where ν, ν' are multiplicity labels distinguishing degenerate J sectors (relevant for $N \geq 3$ spins), and $\langle K, \nu' | \hat{Q}_m | J, \nu \rangle$ is the reduced matrix element, which carries all the dynamics but is independent of M and L . The Clebsch–Gordan coefficient alone imposes two *kinematic* selection rules:

- $\Delta M = k$: the component index $k \in \{-2, -1, 0, +1, +2\}$ sets the change in the magnetic quantum number.
- Triangle rule: $|J - 2| \leq K \leq J + 2$. Combined with the physical constraint $K \leq N/2$, this restricts which J -sectors can be connected.

9.4 Selection rules: two-spin and three-spin cases

Two spins ($J \in \{0, 1\}$). For the singlet $|J = 0\rangle$: the triangle rule requires $|0 - 2| = 2 \leq K$, but $K \leq 1$ for two spins- $\frac{1}{2}$. No K satisfies both constraints simultaneously, so all Clebsch–Gordan coefficients $\langle 0, 0; 2, k | K, L \rangle$ vanish identically for any k . The singlet is dark for *all* components k and *all* spatial indices m —a purely kinematic result requiring no explicit matrix computation. For the triplet $|J = 1\rangle$: the intersection of $\{|1 - 2|, \dots, 3\} = \{1, 2, 3\}$ with $K \leq 1$ gives $K = 1$ only, so the triplet maps only into itself.

Three spins ($J \in \{1/2, 3/2\}$, with the doublet $J = 1/2$ having multiplicity 2).

- From $|J = 1/2\rangle$: triangle rule gives $K \in \{3/2, 5/2\}$; with $K \leq 3/2$ only $K = 3/2$ survives. *Doublet $\rightarrow$ quartet only; doublet $\rightarrow$ doublet transitions are forbidden by the triangle rule.*
- From $|J = 3/2\rangle$: triangle rule gives $K \in \{1/2, 3/2\}$. The quartet can scatter into both doublets and remain in the quartet.

This is the quantitative underpinning of the cross-manifold leakage discussed in Section 6.3: the relaxation superoperator opens quartet $\leftrightarrow$ doublet channels precisely because the triangle rule permits it, whereas within the doublet sector the triangle rule strictly forbids any transition.

9.5 Transition amplitudes: reduced matrix elements via 6j symbols

The selection rules above are kinematic. The actual transition amplitudes are encoded in the reduced matrix element $\langle K, \nu' \| \hat{Q}_m \| J, \nu \rangle$. For a pair operator $T_{2,k}^{(12)}$ acting only on spins 1 and 2, with spin 3 as a spectator, the standard recoupling identity gives

$$\langle J', S'_{12} \| T_2^{(12)} \| J, S_{12} \rangle = (-1)^{S'_{12} + I_3 + J + 2} \sqrt{(2J' + 1)(2J + 1)} \begin{Bmatrix} S_{12} & S'_{12} & 2 \\ J' & J & I_3 \end{Bmatrix} \langle S'_{12} \| T_2^{(12)} \| S_{12} \rangle, \quad (28)$$

where $I_3 = 1/2$ is the spectator spin, the curly bracket is a Wigner 6j symbol, and $\langle S'_{12} \| T_2^{(12)} \| S_{12} \rangle$ is the two-spin reduced matrix element, which vanishes for $S_{12} = 0$ (the pair-singlet dark state, Eq. (16)). For $N = 3$ spins, each of the three pairs contributes a term of the form (28) (with the appropriate spectator spin), and the total reduced matrix element is their sum. Whether contributions add constructively or cancel depends on the molecular geometry (which pairs are dipolar-coupled) and can be evaluated once a consistent S_{12} -labelled basis is fixed.

9.6 Change of jump operator basis

Eq. (26) can be rendered in the IST basis by projecting the transition operators σ_j onto the normalised IST operators $\hat{p}_\delta$

$$\begin{aligned} \dot{\rho} &= -i[H, \rho] - \sum_{i,j} \Gamma_{ij}^{(-)} [\sigma_i^\dagger \sigma_j, \rho] + \sum_{i,j} \Gamma_{ij}^{(+)} \left(-\{\sigma_i^\dagger \sigma_j, \rho\} + 2\sigma_j \rho \sigma_i^\dagger \right) \\ &= -i[H, \rho] - \sum_{ij} \sum_{\mu, \delta} \Gamma_{ij}^{(-)} c_{i,\mu}^* c_{j,\delta} [\hat{p}_\mu^\dagger \hat{p}_\delta, \rho] \\ &\quad + \sum_{i,j} \sum_{\mu, \delta} \Gamma_{ij}^{(+)} c_{i,\mu}^* c_{j,\delta} \left(-\{\hat{p}_\mu^\dagger \hat{p}_\delta, \rho\} + 2\hat{p}_\delta \rho \hat{p}_\mu^\dagger \right) \\ &= -i[H, \rho] - \sum_{\mu, \delta} \Delta_{\mu, \delta}^{(-)} [\hat{p}_\mu^\dagger \hat{p}_\delta, \rho] + \sum_{\mu, \delta} \Delta_{\mu, \delta}^{(+)} \left(-\{\hat{p}_\mu^\dagger \hat{p}_\delta, \rho\} + 2\hat{p}_\delta \rho \hat{p}_\mu^\dagger \right) \end{aligned} \quad (29)$$

where $c_{j,\delta} = \text{Tr}\{\hat{p}_\delta^\dagger \sigma_j\}$ and

$$\begin{aligned} \Delta_{\mu, \delta}^{(-)} &= \sum_{i,j} \Gamma_{ij}^{(-)} c_{i,\mu}^* c_{j,\delta} \\ \Delta_{\mu, \delta}^{(+)} &= \sum_{i,j} \Gamma_{ij}^{(+)} c_{i,\mu}^* c_{j,\delta} \end{aligned}$$

What is the “boddyness” of the predominant channels (δ, μ) ? Can we leverage the symmetry of the transition operators $\sigma_i = |S_1^{(i)}, M_1^{(i)}\rangle \langle S_2^{(i)}, M_2^{(i)}|$ to derive selection rules for

non-vanishing inner products $\langle\langle T_m^{(l)} | \sigma_i \rangle\rangle$ in Liouville space, where $T_m^{(l)}$ is a many-body tensor of rank l ?

Following the intuition that spins decohere on a timescale $\tau_c \ll 1/J_H$, J_H being a characteristic Heisenberg energy scale, the Lindblad operators are expected to preserve the “two-bodiness” of dipolar interactions. The master equation is therefore projected into the two-spin rank-2 IST basis $T_{ij}^{(l=2,m)}$ (indices i, j label a spin pair), corresponding to the change of basis in Eq. (29) with $\{\hat{p}_i\} = \{T_{ij}^{(l=2,m)}\}$.

The five dominant canonical Lindblad operators $\hat{L}_k = \sum_{\delta} U_{\delta k}^* \hat{p}_{\delta}$ are listed in Table 5, decomposed into the normalised two-spin rank-2 IST basis $\{\hat{p}_{\delta}\}$ (unit Hilbert–Schmidt norm).

10 Gemcitabine parameters

Spin-system parameters for Gemcitabine from a Gaussian quantum-chemistry calculation. The 10 nuclei are ordered as $(F_0, F_1, H_0, H_1, H_2, H_3, H_4, H_5, H_6, C_0)$, comprising two ^{19}F , seven ^1H , and one ^{13}C .

10.1 J-coupling matrix (Hz)

Scalar couplings J_{ij} in Hz, upper triangular; dots denote symmetric lower entries.

$$J = \begin{pmatrix} 0 & 175.91 & 0.00 & 0.11 & 0.05 & 6.66 & -0.32 & 0.05 & 11.59 & -226.85 \\ \cdot & 0 & 0.09 & -0.20 & 0.28 & 5.91 & 0.08 & -0.09 & 0.48 & -203.41 \\ \cdot & \cdot & 0 & -3.66 & 0.73 & -0.14 & -0.10 & -0.11 & 0.18 & -0.13 \\ \cdot & \cdot & \cdot & 0 & 2.88 & -0.12 & -0.10 & -0.10 & 0.38 & -0.10 \\ \cdot & \cdot & \cdot & \cdot & 0 & 0.22 & 0.02 & -0.09 & 2.12 & 0.31 \\ \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0.38 & -0.07 & -0.37 & -0.65 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & 2.67 & -0.15 & 0.02 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & -0.14 & -0.08 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 & -2.68 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

10.2 Nuclear coordinates

Nucleus	x (Å)	y (Å)	z (Å)
F_0	0.1666	-1.2783	-1.3875
F_1	0.9661	-2.5748	0.3352
H_0	4.2344	1.2314	-0.5006
H_1	2.7307	1.8998	-1.1779
H_2	2.8928	0.2529	1.3947
H_3	0.4474	-0.6659	1.8321
H_4	-1.7462	-0.8907	2.3837
H_5	-4.1117	-0.4940	1.8959
H_6	2.4649	-0.4748	-1.5086
C_0	0.9133	-1.2851	-0.2046

10.3 Zeeman (chemical shielding) matrices (ppm)

$$\sigma(F_0) = \begin{pmatrix} -130.0601 & -14.9649 & -63.1405 \\ -14.9649 & -215.3146 & 25.9348 \\ -63.1405 & 25.9348 & -160.2068 \end{pmatrix}$$

$$\sigma(F_1) = \begin{pmatrix} -157.2094 & -12.7347 & -67.3755 \\ -12.7347 & -235.8144 & 5.7085 \\ -67.3755 & 5.7085 & -228.8689 \end{pmatrix}$$

$$\sigma(H_0) = \begin{pmatrix} 2.2651 & -0.6354 & 2.4859 \\ -0.6354 & 2.7073 & -2.7546 \\ 2.4859 & -2.7546 & 6.9013 \end{pmatrix}$$

$$\sigma(H_1) = \begin{pmatrix} 7.0348 & -2.4557 & -1.2496 \\ -2.4557 & 3.6480 & -0.3257 \\ -1.2496 & -0.3257 & 1.9779 \end{pmatrix}$$

$$\sigma(H_2) = \begin{pmatrix} 1.5527 & 0.6046 & -2.3124 \\ 0.6046 & 5.3322 & 0.1728 \\ -2.3124 & 0.1728 & 4.7300 \end{pmatrix}$$

$$\sigma(H_3) = \begin{pmatrix} 4.3528 & -1.0174 & 1.5080 \\ -1.0174 & 5.8782 & 1.7827 \\ 1.5080 & 1.7827 & 3.5154 \end{pmatrix}$$

$$\sigma(H_4) = \begin{pmatrix} 3.1215 & 0.1121 & 1.7281 \\ 0.1121 & 9.5089 & 0.5658 \\ 1.7281 & 0.5658 & 7.4617 \end{pmatrix}$$

$$\sigma(H_5) = \begin{pmatrix} 2.5152 & 0.8210 & -0.8704 \\ 0.8210 & 6.6786 & 1.2727 \\ -0.8704 & 1.2727 & 4.7920 \end{pmatrix}$$

$$\sigma(H_6) = \begin{pmatrix} 4.4701 & 0.6055 & 0.2616 \\ 0.6055 & 4.4715 & 1.0493 \\ 0.2616 & 1.0493 & 2.7238 \end{pmatrix}$$

$$\sigma(C_0) = \begin{pmatrix} 241.3681 & -1.4291 & -6.1424 \\ -1.4291 & 245.7345 & 3.0563 \\ -6.1424 & 3.0563 & 250.0215 \end{pmatrix}$$

10.4 Change of jump operator basis

Eq. (26) can be rendered in the IST basis by projecting the transition operators σ_j onto the normalised IST operators $\hat{p}_\delta$

$$\begin{aligned}
\dot{\rho} &= -i[H, \rho] - \sum_{i,j} \Gamma_{ij}^{(-)} [\sigma_i^\dagger \sigma_j, \rho] + \sum_{i,j} \Gamma_{ij}^{(+)} \left(-\{\sigma_i^\dagger \sigma_j, \rho\} + 2\sigma_j \rho \sigma_i^\dagger \right) \\
&= -i[H, \rho] - \sum_{ij} \sum_{\mu,\delta} \Gamma_{ij}^{(-)} c_{i,\mu}^* c_{j,\delta} [\hat{p}_\mu^\dagger \hat{p}_\delta, \rho] \\
&\quad + \sum_{ij} \sum_{\mu,\delta} \Gamma_{ij}^{(+)} c_{i,\mu}^* c_{j,\delta} \left(-\{\hat{p}_\mu^\dagger \hat{p}_\delta, \rho\} + 2\hat{p}_\delta \rho \hat{p}_\mu^\dagger \right) \\
&= -i[H, \rho] - \sum_{\mu,\delta} \Delta_{\mu,\delta}^{(-)} [\hat{p}_\mu^\dagger \hat{p}_\delta, \rho] + \sum_{\mu,\delta} \Delta_{\mu,\delta}^{(+)} \left(-\{\hat{p}_\mu^\dagger \hat{p}_\delta, \rho\} + 2\hat{p}_\delta \rho \hat{p}_\mu^\dagger \right) \tag{30}
\end{aligned}$$

where $c_{j,\delta} = \text{Tr}\{\hat{p}_\delta^\dagger \sigma_j\}$ and

$$\begin{aligned}
\Delta_{\mu,\delta}^{(-)} &= \sum_{ij} \Gamma_{ij}^{(-)} c_{i,\mu}^* c_{j,\delta} \\
\Delta_{\mu,\delta}^{(+)} &= \sum_{i,j} \Gamma_{ij}^{(+)} c_{i,\mu}^* c_{j,\delta}
\end{aligned}$$

What is the “boddyness” of the predominant channels (δ, μ) ? Can we leverage the symmetry of the transition operators $\sigma_i = |S_1^{(i)}, M_1^{(i)}\rangle \langle S_2^{(i)}, M_2^{(i)}|$ to derive selection rules for non-vanishing inner products $\langle \langle T_m^{(l)} | \sigma_i \rangle \rangle$ in Liouville space, where $T_m^{(l)}$ is a many-body tensor of rank l ?

Following the intuition that spins decohere on a timescale $\tau_c \ll 1/J_H$, J_H being a characteristic Heisenberg energy scale, the Lindblad operators are expected to preserve the “two-bodiness” of dipolar interactions. The master equation is therefore projected into the two-spin rank-2 IST basis $T_{ij}^{(l=2,m)}$ (indices i, j label a spin pair), corresponding to the change of basis in Eq. (29) with $\{\hat{p}_i\} = \{T_{ij}^{(l=2,m)}\}$.

The five dominant canonical Lindblad operators $\hat{L}_k = \sum_\delta U_{\delta k}^* \hat{p}_\delta$ are listed in Table 5, decomposed into the normalised two-spin rank-2 IST basis $\{\hat{p}_\delta\}$ (unit Hilbert–Schmidt norm).

(a) $^{19}\text{F}-^{13}\text{C}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.9971	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	-0.0000	+0.0914	+0.0119	-0.1234	+0.9852
$\hat{L}_3$	+0.0000	-0.8828	-0.2426	+0.3724	+0.1315
$\hat{L}_4$	-0.0000	+0.1146	-0.9353	-0.3236	-0.0398
$\hat{L}_5$	+0.0000	-0.4397	+0.2458	-0.8577	-0.0697
(b) $^{19}\text{F}-^1\text{H}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.0708	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	+0.0000	+0.0065	+0.0008	-0.0088	+0.0700
$\hat{L}_3$	+0.0000	-0.0627	-0.0172	+0.0265	+0.0093
$\hat{L}_4$	+0.0000	+0.0081	-0.0665	-0.0230	-0.0028
$\hat{L}_5$	-0.0000	-0.0312	+0.0175	-0.0609	-0.0050
(c) $^{13}\text{C}-^1\text{H}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.0270	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	+0.0000	+0.0025	+0.0003	-0.0033	+0.0267
$\hat{L}_3$	-0.0000	-0.0239	-0.0066	+0.0101	+0.0036
$\hat{L}_4$	-0.0000	+0.0031	-0.0254	-0.0088	-0.0011
$\hat{L}_5$	+0.0000	-0.0119	+0.0067	-0.0233	-0.0019

Table 5: Decomposition of the five dominant canonical Lindblad operators $\hat{L}_k$ in the normalised two-spin rank-2 IST basis $\hat{p}_\delta = T_\delta^{(2,k)} / \|T_\delta^{(2,k)}\|_{\text{HS}}$. Sub-tables (a), (b), (c) list coefficients $U_{\delta k}^*$ for the $^{19}\text{F}-^{13}\text{C}$, $^{19}\text{F}-^1\text{H}$, and $^{13}\text{C}-^1\text{H}$ pairs respectively. Dephasing rate: $2d_k/(2\pi) = 10616 \text{ Hz}$.

(a) $^{19}\text{F}-^{13}\text{C}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.9971	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	-0.0000	+0.0914	+0.0119	-0.1234	+0.9852
$\hat{L}_3$	+0.0000	-0.8828	-0.2426	+0.3724	+0.1315
$\hat{L}_4$	-0.0000	+0.1146	-0.9353	-0.3236	-0.0398
$\hat{L}_5$	+0.0000	-0.4397	+0.2458	-0.8577	-0.0697
(b) $^{19}\text{F}-^1\text{H}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.0708	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	+0.0000	+0.0065	+0.0008	-0.0088	+0.0700
$\hat{L}_3$	+0.0000	-0.0627	-0.0172	+0.0265	+0.0093
$\hat{L}_4$	+0.0000	+0.0081	-0.0665	-0.0230	-0.0028
$\hat{L}_5$	-0.0000	-0.0312	+0.0175	-0.0609	-0.0050
(c) $^{13}\text{C}-^1\text{H}$	$k = -2$	$k = -1$	$k = 0$	$k = +1$	$k = +2$
$\hat{L}_1$	+0.0270	+0.0000	+0.0000	+0.0000	+0.0000
$\hat{L}_2$	+0.0000	+0.0025	+0.0003	-0.0033	+0.0267
$\hat{L}_3$	-0.0000	-0.0239	-0.0066	+0.0101	+0.0036
$\hat{L}_4$	-0.0000	+0.0031	-0.0254	-0.0088	-0.0011
$\hat{L}_5$	+0.0000	-0.0119	+0.0067	-0.0233	-0.0019

Table 6: Decomposition of the five dominant canonical Lindblad operators $\hat{L}_k$ in the normalised two-spin rank-2 IST basis $\hat{p}_\delta = T_\delta^{(2,k)} / \|T_\delta^{(2,k)}\|_{\text{HS}}$. Sub-tables (a), (b), (c) list coefficients $U_{\delta k}^*$ for the $^{19}\text{F}-^{13}\text{C}$, $^{19}\text{F}-^1\text{H}$, and $^{13}\text{C}-^1\text{H}$ pairs respectively. Dephasing rate: $2d_k/(2\pi) = 10616 \text{ Hz}$.